Mineralization in the Arizona Context

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Introduction

Concerns over anthropogenic climate change will drive the development and deployment of negative emission technologies (Shukla *et al.* 315; Davis *et al.*, secs. 32–24). Among these, direct air capture combined with carbon storage holds great promise because direct air capture is inherently scalable and allows existing infrastructure to live out its useful life (Committee on Developing a Research Agenda for Carbon Dioxide Removal and Reliable Sequestration *et al.* 189). Its operational scale can be adapted to meet local requirements. However, at present it is far too expensive to see widespread deployment (Lackner and Azarabadi 8202).

In many regions of the world, and especially in the US, geological storage is a preferred approach for carbon storage (National Acadamies 14). However, it is not feasible everywhere. Arizona lacks widespread deep sedimentary structures that are suitable for conventional geologic storage. Instead, there are widespread deposits of rocks suitable for mineral sequestration, either *in situ* in underground formations or *ex situ* processing minable deposits (Sjolund *et al.* 2). Unlike in basalt regions in Iceland or the Columbia River Basin, deposits in Arizona are typically localized and individually small (Amiotte Suchet *et al.* 3; Sjolund *et al.* 2). Nevertheless, these deposits represent an economic opportunity for cost-effective carbon dioxide removal. However, they require flexible and modular technologies for carbon delivery as large permanent infrastructures for carbon dioxide delivery could easily outlast the capacity of the reservoir. Direct air capture offers a modular approach that can adapt to local demand patterns. Furthermore, the integration of direct air capture with mineral sequestration offers an opportunity to simplify and streamline both technologies.

This report summarizes our work on connecting direct air capture technologies with mineralization within the Arizona context. The report is part of a larger study to catalog and explore mineralizable mafic rock resources throughout the state and to characterize their reactivity for mineralization.

In chapter two of the report, we revisit mineralization processes with a focus on the idiosyncrasies of Arizona mineral deposits. The chapter outlines a wide range of technology options and how to integrate them with direct air capture again considering the Arizona situation. It then by necessity has to make a choice of technologies for further study and develops a specific process model for what at present is considered the most promising approach for the immediate future. As a result, maturity of the process was a major consideration of choosing a technology. However, some of the less mature approaches may have long-term viability and should be considered for further development.

Chapter three creates a process model of the particular technology choice. Since none of the strongly integrated systems that tightly couple direct air capture to mineralization was mature enough for further analysis the deviation from standard air capture was minimal and most of the work focused on the carbonation of mafic minerals in the presence of bicarbonate solutions. The model resulted in mass and heat balances that in chapter four were the basis for a high-level techno-economic and life cycle analysis.

The conclusion in chapter 5 seeks to evaluate these results and explore a way forward for the technology. The main hurdle to be overcome is the high cost in preparing the resource minerals for conversion to carbonates.

Process Ideation

Carbon mineralization as a means to sequester carbon dioxide has been grounded by and ultimately made possible because of work in the geoscience community. It has been understood since the work of Harold Urey that siliceous rocks play a key role in mediating the mobile carbon pool comprising the atmosphere, biosphere and oceans (Urey). The mobile carbon pool in turn controls global climate. This has been expanded into foundational work with Berner's work developing the carbonate-silicate geochemical cycle (Berner *et al.*). Berner's work established that these fluxes are significant over geological timescales, with fluxes due to silicate weathering being in the range of 0.5 of gigatonnes per year, even though the immediate impact is small (Berner *et al.* 649). An understanding of this underlying geological process naturally led to the question if this process could be accelerated to mitigate anthropogenic climate change (Seifritz).

The first intensive study on intentionally mineralizing carbon dioxide to prevent global warming was performed by H.E. Dunsmore. In this paper he described the need for mineralization to balance the emissions of anthropogenic climate change with permanent sequestration (Dunsmore 567). The core of his plan was to exploit high calcium and magnesium content chloride brines, to produce hydrochloric acid and precipitate carbonates (Dunsmore 571). This approach would see the need to sequester large quantities of HCl in silicate materials (Dunsmore 571). This ultimately argues for a vison of carbon mineralization that causes the management of excess atmospheric carbon shifted to being one of managing disposal of concentrated acids.

The work by Lackner, Wendt and later joined by Goff starting in 1995 were the first significant effort to explore carbon sequestration through mineralization of carbonates directly with calcium and magnesium rich silicate rocks. Two approaches were considered, direct carbonation, and indirect acid leaching processes (K. S. Lackner, Wendt, *et al.* 1162). The later approach is conceptually similar to Dunsmore's approach, although the process envisioned is not a net consumer of chloride ions (K. S. Lackner, Wendt, *et al.* 1163). The direct mineralization approach where CO_2 directly reacts with calcium and magnesium rich rock is made possible by the favorable thermodynamics across a range of temperatures (K. S. Lackner, Wendt, *et al.* 1158). Since the free energy of calcite and magnesite, the principal calcium and magnesium carbonate minerals, are dependent on the partial pressure of CO_2 over the reaction carbonate reactions have a wider thermodynamic window at elevated pressures. Increasing the pressure of CO_2 increases the chemical driving force and allows the reaction to occur at higher temperatures (K. S. Lackner, Wendt, *et al.* 1158). The ability to react at higher temperatures was seen as important to achieve favorable reaction rates (K. S. Lackner, Wendt, *et al.* 1162). The envisioned process was for sequestering carbon dioxide from a coal fired power plant that would have needed to be transported to the mine site for mineralization (K. S. Lackner, Wendt, *et al.* 1164). In further work by Goff and Lackner the envisioned process would see the potential for coproduction of valuable byproducts (Goff and Lackner).

Building off the work of Lackner *et al.* there was a sustained period of investment at multiple national labs in the United States, and with sustained industry investment in the topic. Albany Research Center lead a key project in establishing the behavior of direct mineralization systems (O'Connor, Dahlin, Nilsen, Walters, *et al.*). This work has laid the groundwork for the industrial scale pilot plants and has been the dominant framework within which *ex situ* mineralization has been considered. However, other approaches are also worth considering. For example, Wendt and Lackner studied the thermodynamics of magnesium chloride melts in contact with CO_2 and magnesium silicates and suggested direct carbonation of olivine and serpentine in the presence of high pressure CO_2 inside a magnesium melt (Wendt *et al.*).

The study of industrial ex situ mineralization virtually always assumed large scale mining operations (K. S. Lackner, Wendt, *et al.* 1164; Gerdemann *et al.* 2590). Gigawatt coal plants producing about 7 megatonnes of CO_2 per year require large mining operations. This has limited the sites available to few ultramafic surface deposits capable of storing decades worth of emissions from one or more powerplants. Direct air capture has no such constraints (K. Lackner et al.; K. S. Lackner, Brennan, et al.). Individual capture modules are unlikely to operate at the megaton a year scale. Indeed, they could be small enough to be relocated during their operating lifespan if locally available storage capacity was depleted. The more granular nature of direct air capture allows for the exploitation of smaller mineral deposits that otherwise would have been unable to meet demand. This necessitates a new analysis of available resources that considers deposits that are too small to sustain high carbon disposal rates for decades. They may even be too small to reach disposal rates commensurate with a large power plant. On the other hand, as this study shows, in aggregate these resources make a substantial contribution to the overall storage capacity.

Most direct air capture technologies aim to produce pipeline ready CO_2 (Azarabadi and Lackner, "A Sorbent-Focused Techno-Economic Analysis of Direct Air Capture"; Azarabadi and Lackner, "Postcombustion Capture or Direct Air Capture in Decarbonizing US Natural Gas Power?"; Morgan *et al.*). Their technical requirements are driven by pipeline constraints. Pipelines require high pressures and high purities. Especially water vapor and oxygen must be tightly controlled (Shirley and Myles). Air capture systems that generate high pressure CO_2 while relying on sorbents with close to the minimum required binding energy will nearly always involve multiple stages of raising CO_2 concentrations and pressures. In the absence of pipeline requirements, these constraints may change to match the requirements of the end consumer. For mineralization this may relax pressure and purity constraints. However, the requirements will vary dramatically with the choice of mineralization technique. High pressure mineralization processes have similar requirements to conventional direct air capture with regard to pressure, but purity constraints could still be far more relaxed. However, corrosion issues arising in high pressure CO_2 with admixtures of water and oxygen will likely maintain a strong incentive to limit their concentrations even within the DAC process itself. On the other hand, for mineralization processes that do not require high pressure, the relaxed technical requirements could greatly simplify direct air capture.

A very different direct air capture system takes advantage of the fact many mineralization processes do not need gaseous CO_2 but instead utilize carbonate or bicarbonate ions. Carbonate/bicarcarbonate brines can be produced with a single stage direct air capture system. For example, moisture swing sorbents, can readily produce CO_2 or carbonate and bicarbonates at fugacities of a few kilopascals and thus can produce brines ready for mineralization (Wang *et al.*; Lackner). It is conceivable that the brines used to react with the minerals could also be used for direct air capture. However, the reaction kinetics of such brines are typically too slow to be used directly as a sorbent. They also may contain too many other salts to be used in direct contact with direct-air-capture sorbents. Therefore, typically such a system would involve a gas interface as a barrier to prevent salts from crossing between the mineral reactor and the direct air capture system. There is plenty of potential for further integration. Unfortunately, this space has not been well explored and our focus for this report will remain on systems with higher technical readiness levels.

For any carbon sequestration to be useful it must satisfy certain performance standards and at the same time be affordable. *Ex situ* mineralization competes with many other sequestration technologies including biomass storage, and geological sequestration, which includes *in situ* mineral sequestration (Matter *et al.*). What mineral sequestration in general has to offer is an extremely high potential for permanence. The probability that these minerals decompose within climate relevant time scales, when sequestered at scale, is extremely small. *Ex situ* mineralization in addition allows for an accurate assessment of how much carbon has been mineralized, therefore simplifying the process of certifying carbon removal. However, *ex situ* mineral sequestration is more challenged when considering cost and its social and environmental footprint. Given the nascent state of sequestration technologies, how much pain will be inflicted by climate change and how societal attitudes will change with increased understanding of carbon management, and with other evolving societal priorities, is impossible to predict the ultimate tradeoffs between these three categories of durability, economics and socio-ecological impact. Nevertheless, it is necessary to specify reasonable requirements to choose among the various options outlined above.

It is likely that mineral sequestration will command a premium over less durable sequestration methodologies, especially bio-based sequestration approaches. Another reason for fetching a premium is the high degree of accountability for the carbon. The association with mining would tend to lower the value of *ex situ* mineral sequestration (Gobster and Smardon). However, mineral sequestration may be performed in conjunction with other mining activities that create value and it even may generate products which by themselves have value. In geological sequestration transport of CO_2 to the storage well frequently represents a large fraction of the overall cost. Hence, mineral sequestration may be competitive in some situations by minimizing transport distances. In our case by coupling mineralization with air capture these transport costs have been completely eliminated but have effectively been replaced by the cost of direct air capture. Because direct air capture in most cases will exceed the cost of pipeline transport, we cannot give credit for the avoidance of CO_2 transport (Azarabadi and Lackner, "A Sorbent-Focused Techno-Economic Analysis of Direct Air Capture"; Committee on Developing a Research Agenda for Carbon Dioxide Removal and Reliable Sequestration *et al.*). Here we assume that economic viability ultimately requires mineralization costs below $100/tonne after taking credit for other economic drivers for utilizing a particular mine.

In the context of this report, one must consider the idiosyncrasies of mineral deposits specific to Arizona. In contrast to large basalt flows which have been explored elsewhere, or vast ophiolite deposits suitable minerals in Arizona are often found in highly localized formations (Callow *et al.*; McGrail *et al.*; Goldberg *et al.*; Goldberg and Slagle). The difference between these large reservoirs and the much smaller ones considered in Arizona requires the evaluation of a broader portfolio of technologies than have been previously assed for landscape defining formations. Geography and climate also play an important role; in Arizona it is particularly important to limit water requirements. Given the critical nature of water in arid environments like Arizona, water limitations often go beyond simple cost concerns, but also raise larger societal concerns. Thus, additional water use may lack the societal license necessary for operation, even if water would not be a cost issue for the new operation available (Cesar and Jhony; Dolan *et al.*; Shirley and Myles; Trelease).

Scoping Integrated Mineralization and Direct Air Capture (DAC)

A systematic process was adopted to scope out a tightly coupled DAC and mineralization system informed both by historic research and by the ongoing characterization and resource mapping performed as part of this project. The development process consists of three phases: documentation of the scoping study, preliminary system design, and a second iteration on the system design that settles on some of the specifics. Design space scoping was performed through a formal process to create a decision tree chart documenting the key choices made in creating the system's basic design. The scoping documentation process was informed by previous work conducted on mineralization for carbon sequestration and revealed key constraints and raised important questions related to the characterization of the mineral resource. Many of the answers to these questions will need to be informed by additional studies and future experimental results.

The second step is the preliminary system design. This resulted in a traditional process sheet describing the process concept, the unit processes involved, and quantifying the material flows through the system. The preliminary system model was in part informed by experimental results for this project. It raised important questions about controlling the formation of unwanted by-products (Sjolund *et al.*). The work also has also begun to capture important process steps with advanced modeling techniques using modern simulation systems, like object oriented Modelica tools. The last step involves a more detailed design. However, as we approach this stage it became clear that the remaining uncertainties are still too large to complete this design.

Conceptual Process Design

In preparation for selecting among different process designs, a comprehensive option space analysis has been performed and documented. The analysis considers a wide range of possible approaches to Direct Air Capture and Mineralization. The design spaces have been documented in decision tree charts describing the challenges faced and the selection requirements to eventually arrive at a final design. The ultimate choice typically depends on conditions at the selected mineral deposit. The final design may still change as more data further constrain the various options.

The decision tree is split into two individual charts. The design decisions for the DAC system in the second chart (Fig. 2) are strongly influenced by the upstream design decisions for the mineralization system outlined in figure 1. The ultimate choices made in the mineralization process led to the initial choices that will have to be made in figure 2. Some processes will require concentrated (or even pressurized) CO_2, whereas other process designs can utilize CO_2 at low partial pressures. Once a mineralization process has been selected, the DAC method, which is strongly affected by the specification of the concentration and state of the CO_2, will then be identified and selected. The optimal choice is driven by the requirements of the mineralization process. Early modeling and scoping analyses suggest that the in the coupled system the DAC sub-system is easier to adapt to the needs of the mineralization sub-system than for mineralization system to adapt to the limits and constraints of a specific DAC technology.

During the development of the technological option space for Direct Air Capture and Mineralization, minimizing the use of fresh water throughout the process emerged as a key requirement. The focus on arid regions in Arizona further emphasized this requirement. Minimizing freshwater requirements has led us to favor options that use saline brines to bring CO_2 in contact with the rock. However, ion exchange resins, which are strong sorbent contenders for a moisture driven DAC process, cannot tolerate direct contact with saline brines that contain anions other than hydroxide, carbonate or bicarbonate. To a lesser extent these impurities also pose a problem for polymer-based amine sorbents that are regenerated in a thermal swing. Approaches that can use saline brines for mineralization but avoid direct contact of the brine with the DAC sorbent during regeneration are therefore critical.

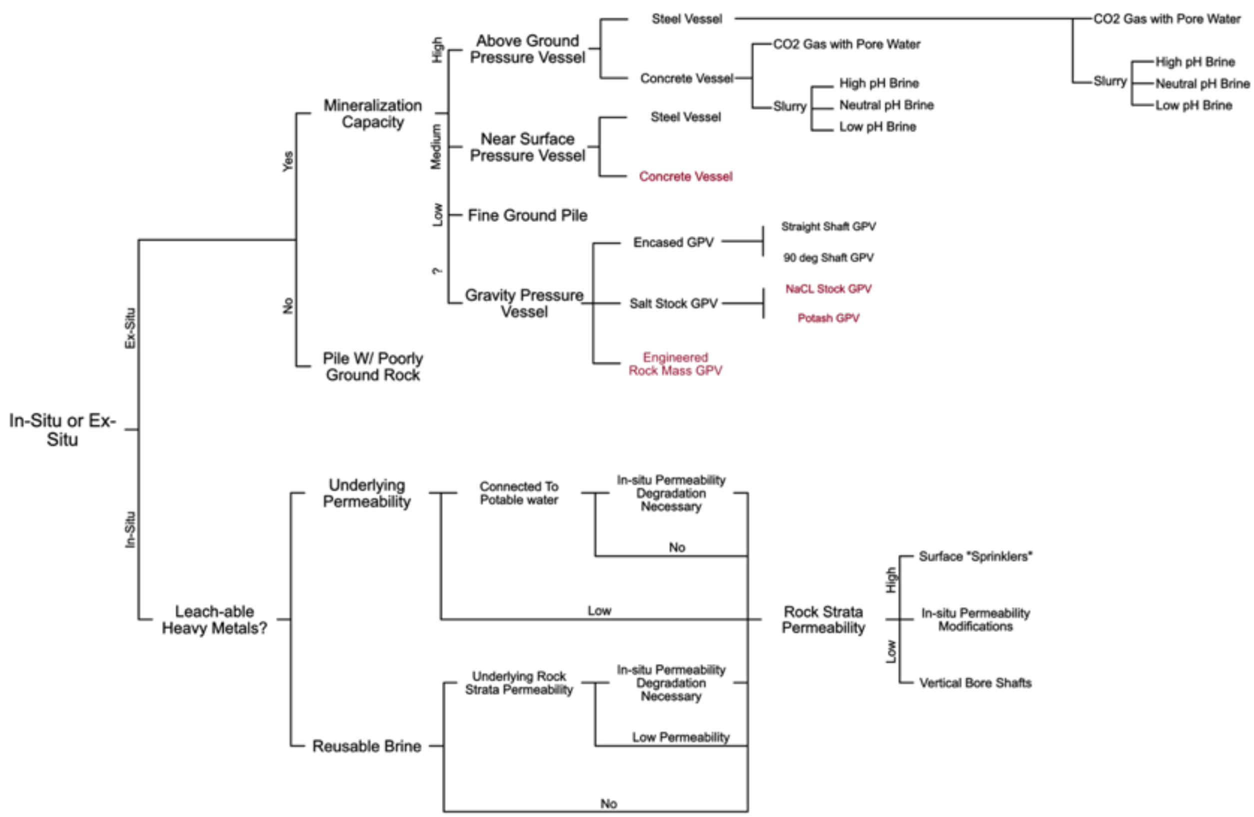

Figure 1 Mineralization Decision Tree Chart
The chart is read from left to right with the terminating branches on the right describing the final form of the mineralization system. Each system is characterized by the binary choices made in the decision tree. The decisions depend on circumstances at the particular site of the mineral and on the evolving understanding of different process designs. Each decision depends on the answer to a question or it represents an engineering choice that affects the design for the mineralization system. By forcing answers to the underlying questions, controlling factors for the design of the system can be readily identified. For some nodes of the decision tree there is not an explicit question asked as they instead involve design trade-offs that must be understood at a deeper level of enginering. For example, after choosing "Above Ground Pressure Vessel" two types of reactor materials are identified and all suport similar aproaches for bring CO_2 into contact with the rock. The design choice will come down to a variety of factors including cost, overall pressure required, material compatibility, and manufacturability which is not yet entirely constrained at this level of analysis. Approaches marked with maroon text are considered to be potentially dissruptive technologies for mineralization. They could rapidly alter the cost curve for construction but are at a very low technolog readiness level. Because they inject large uncertainties and likely large development costs into such commercialization effort, these potentially disruptive technologies were not chosen for a more in-depth study as part of this project.

Even saline brines tend to suffer from significant water losses due to evaporation either during contact with the mineral material or due to drying out of the DAC sorbent during CO_2 collection from the air. One approach to minimize evaporative losses is to use hygroscopic brines that in some cases can extract water vapor from the atmosphere. Phosphates were identified as potential sorbents or additives to the brine used in mineralization, as they are highly hygroscopic which could be leveraged for use in a desert environment.

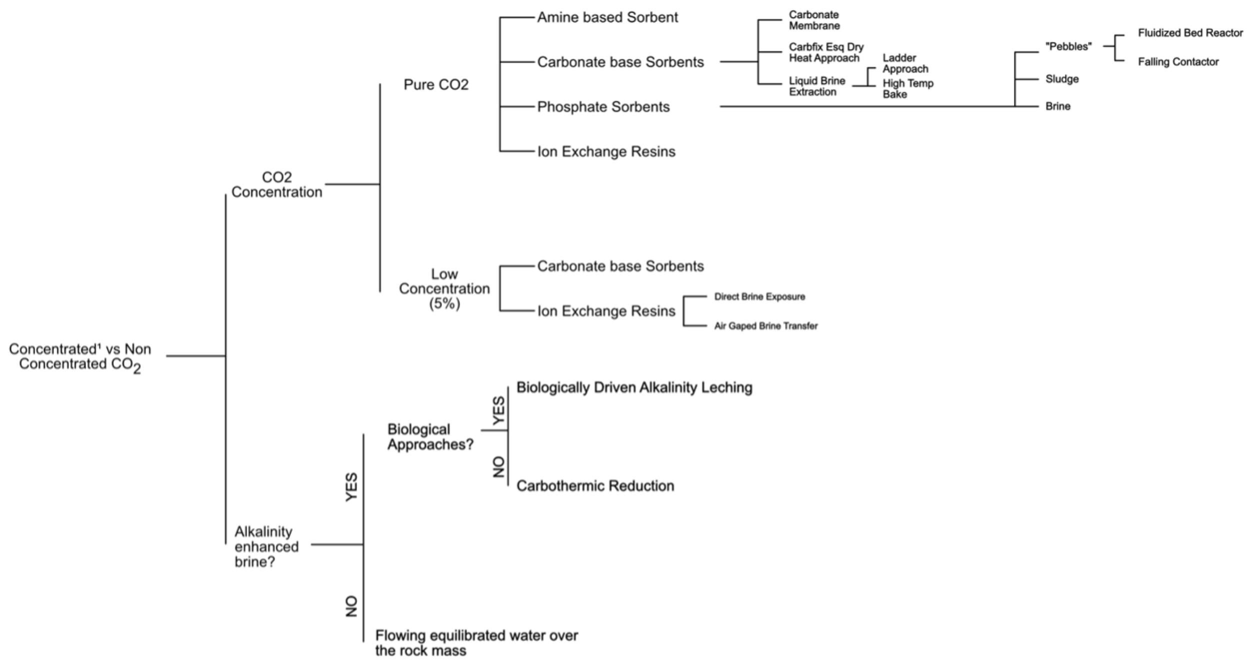

Figure 2 Direct Air Capture Decision Tree
The DAC decision tree follows a similar logic to that of the mineralization decidision tree. Many of the binary choices in the DAC decision tree depend on the choices made in the mineralization decision tree. Mineralization can potentially offer far more relaxed product specifications for the captured CO_2 than is normally the case for DAC systems. Low partial pressures of CO_2, or in the case of aqueous solutions, low fugacities of CO_2, are acceptable for a number of mineralization technologies of interest. Rather than producing a stream of concentrated and pressurized CO_2 that lives up to enhanced oil requirement, or pipeline requirement, in the case of on-site mineralization of CO_2 mixed with air and partical pressures in the kPa range may prove sufficient. Another avenue of interest may involve aqueous bicarbonate solutions with CO_2 fugacities in the kPa range. Central issues then revolve of the management of other impurities that could affect the performance of the DAC system.

Beyond the limitations on fresh water, the decision trees for technology choices identified additional key factors in developing potential commercial approaches to exploiting scoria deposits for mineralization. However, design decisions depend on material properties and reaction kinetics which are only in the early stages of exploration. An important result from other parts of this project is that many of the mineral resources including scoria can be effectively carbonated at low pressures and low temperatures (Sjolund *et al.*). This, however, requires designs that allow for slow reaction times. Key factors that shape design decisions included the susceptibility of DAC sorbents to damage from exposure to rock materials, the need to minimize the cost of reactor vessels that can manage slow reaction rates, and the identification of processes that do not require continuous power input. The latter is important, because most renewable energy sources operate intermittently, and storage costs need to be kept at a minimum. Grid energy which comprises a large amount of fossil energy is not only expensive, it also creates a high burden of indirect emissions. Design development focused on managing these constraints while utilizing the initial experimental results from collected scoria samples.

Comparison to Previous Work

A first detailed analysis of *ex situ* mineralization appeared in 1995 (K. S. Lackner, Wendt, *et al.*; K. S. Lackner, Butt, *et al.*). The paper discussed the potential resource base, underlying chemistry and thermodynamic favorability, and proposed two approaches to mineralization, a direct reaction approach involving either a direct gas-solid reaction or a reaction in an aqueous environment, and a hot HCl leaching process (Wendt *et al.*; K. S. Lackner, Wendt, *et al.*). Studies at Los Alamos further explored resource availability and process design for mineral sequestration. Similar work was performed at Ohio State (Alissa Park). Further work at Arizona State University by Mike McKelvy, Andrew Chizmesia, *et al.* explored the reaction barriers in the dissolution of olivine and serpentines (Bearat *et al.*). Mazzotti explored the reaction kinetics (Prigiobbe *et al.*; Hänchen *et al.*), while the Albany research center focused on process designs of heat-treated serpentinites (O'Connor, Dahlin, Nilsen, Rush, *et al.*; Gerdemann *et al.*; O'Connor, Dahlin, Nilsen, Walters, *et al.*). Krevor combined work on resource mapping and reaction kinetics of serpentinites in various salt that could accelerate magnesium leaching while eliminating the silica crust forming around a dissolving particle (Krevor and Lackner; Krevor *et al.*). This work was extended by researchers at Orica leading to an Australian spinout company that is currently operating a serpentine based mineralization process. While Rashid *et al.* argued to remove this barrier to effective mineralization to perform the mineralization action with concurrent grinding of the material. Other work by Greg Dipple and Georges Beaudoin started with the goal of carbonating mine tailings. It appears many of these ultra mafic mine deposits contain a sufficient amount of brucite to spontaneously carbonate (Power *et al.*; Wilson *et al.*; Harrison *et al.*). However in the case of Georges Beaudoin the carbonation of serpentinites is likely to play a role as well (Pronost *et al.*; Lechat *et al.*; Gras *et al.*; Assima, Larachi, Molson, *et al.*; Assima, Larachi, Beaudoin, *et al.*). Their work points to the reverse heap leaching we apply here. An even slower set of processes is based on accelerated mineral weathering is the extreme endpoint of this approach where mafic materials are spread onto shorelines or agricultural fields to mineralize CO_2 directly from the air (van Herk *et al.*; ten Berge *et al.*; Schuiling and Krijgsman). Paul Knops further developed a number of practical implementations of accelerated weathering, including the use of olivine for packings on railroad tracks (Vink and Knops).

Narrowing Options for Immediate Deployment

Looking through the various options developed in the figures above; we note that only a small subset has been carefully studied. Substantial work would be needed to contrast and compare all the options laid out in these decision trees. Among the more exotic options pressure containment through deep underground structures offers an innovative and potentially cost-effective means of mineralization. In effect it allows for the benefits of high temperature and pressure processes without the high cost of conventional engineered vessels. The round trip of the fluid from ambient pressures at the surface to the high pressures naturally maintained deep below ground is energetically very cheap. All that needs to be provided is the force required to overcome friction. The liquid column generates its own hydrostatic head and pouring in liquid at one end of a U syphon guarantees outflow on the other. Given the long pipes connecting the underground reaction chamber with the surface one can consider effective heat exchange methodologies that take advantage of the relatively long travel time. Moreover, the relatively low capital cost associated with reaching pressure and temperature makes it possible to consider much longer reaction times than otherwise would be economical.

Unfortunately, these technologies are at best in the early development stage and thus not available for immediate deployment. The second criterion for us to advance a technology to the process development stage is a minimum level of technology maturity. While it is generally difficult to assign accurate technology readiness levels to a wide range of technologies, we require that processes have already been demonstrated under realistic and relevant conditions to be considered in the next phase of this project. Typically, this corresponds to TRL 6 (Department of Energy). Because of this requirement, our immediate focus is on simple reaction designs with small vessels that do not require pressurization. This has ruled out technologies that rely on pressurized CO_2 for the formation of mineral carbonates. Instead, we utilize bicarbonate brines to carry out carbonation. Pressurization may still occur in the formation of these brines, but the brine itself need not be pressurized when brought in contact with the mineral resource.

Bicarbonates form if CO_2 comes in contact with aqueous solutions. For solutions with high alkalinity the capacity of the solution to hold CO_2 can be quite substantial, and some percentage of this dissolved carbon can be utilized in the carbonation reaction, which will end once the fugacity of the carrier fluid has dropped below the thermodynamic threshold required for successful carbonation.

The move to bicarbonate makes it possible to simplify the DAC system as it eliminates the need for high purity and pressurization. This opens the door for a wider class of DAC systems not all of which could be considered in the context of producing pipeline ready pressurized CO_2. On the other hand, we found that deeper integration of the DAC system with the mineralization system directly collides with our requirement for a reasonable technology readiness level. Consequently, our entire focus in process design is on the more novel mineralization process and we essentially assume that direct air capture can deliver a bicarbonate brine to this process. However, it should be noted that the cost of DAC technologies also needs to be reduced for this combination to become economically feasible (Lackner and Azarabadi).

This particular set of processes is very dependent on the specific properties of the raw material. We therefore had to test the suitability of minerals found in various deposits in Arizona for this technical approach to mineral sequestration. One issue that arose is the high heterogeneity of the response of scoria to mineralization. Very different mineralization properties were found for materials from adjacent sampling sites. The exact cause of these differences remains to be ferreted out. In short, the mild conditions under which carbonation is accomplished makes the process quite sensitive to fine details in the mineralogy. Findings along these lines have been published by others on this project (Sjolund *et al.*).

Process Design

The demonstrated effectiveness of low temperature and pressure reactions for the scoria deposits under consideration for development indicated process designs that could exploit DAC systems that can only reach fugacities of CO_2 in the kPa range but allow for long reaction times under mild conditions of pressure and temperature. The basis for these designs is the successful demonstration of carbonation of finely ground mineral samples from targeted source sites. The timescales for completing the mineralization for such processes is measured in weeks to months.

Long reaction times imply large volume of mineral material in the reaction chamber. This in turn demands that the capital cost of the reactors has to be very low. Fortunately, near ambient conditions allow for such designs.

Analysis of the decision trees resulted in a preliminary design of an *ex situ* approach with above-ground (or on the ground) reactors operating at near ambient conditions. The foundation for this decision was based on the high mineralization capacities observed in early mineralization experiments on samples collected in the field. Decisions on the reactor design and CO_2 transport mechanism still allows for a number of options even though high pH brines and CO_2 gas systems were selected to be advanced further in the preliminary system design. In keeping with the use of low-pressure CO_2 and the wide variety of systems that can provide it, the DAC system design may take the form of any of the many branches of the Low CO_2 Concentration part of the decision tree. The attractiveness of moisture swing sorbents for these applications has focused the design on systems that assure that the sorbents are not directly exposed to the rocks and the brines that touch them, by introducing an "air gap" in the system. Since many other sorbents are also sensitive to the impurities present in the mineral material and in the brine that contacts them, air gaps generally provide a safer design albeit more complex design.

The use of mild conditions results in reaction times longer than a week. This drives the need for large volumes of rocks to be managed and exposed to CO_2 at any given point in time. For this reason, the developed processes must minimize the capital costs associated with containing the rock which is made easier by minimizing pressures and by the design of simple flow-through systems. Costs are minimized by containing the ground up rock material in a simple pile that is exposed to CO_2 or a carbonate brine. The resulting approach is similar to resource extraction in heap leaching where the rocks subjected to leaching may require weeks or months of exposure to the leaching solution. Mineralization which binds CO_2 to the mineral, can be envisioned as reverse leaching. Rather than extracting a value from the mineral and adding it to the solution, a solution loaded with dissolved inorganic carbon relinquishes its carbon content which binds the carbonic acid to the mineral matrix. Conceiving of the system as reverse leaching, the design space for reactors and processes is expanded beyond past work of *ex situ* mineralization systems that require large and expensive pressure vessels that are not affordable at the volumes required here. This still leaves two options. One is to have the brine absorb CO_2 outside the mineral pile and thus represent the only moving component in the system. The other option is to deliver CO_2 from the bottom of the pile with the brine mainly trickling down providing a wet interface between the mineral and the gas phase.

From a process perspective the reactor could be open to the environment or protected from contact with the air through simple barriers like geo-textiles or a cover layer on top. Carbonated liquid flows from the top to the bottom of the pile, either carrying its entire carbon content with it or making contact with gas-phase CO_2 that percolates upward through the pile. The low operating pressures and ambient temperatures allow for designs which can hold large quantities of minerals at low cost. Such processes favor designs work with simple DAC systems that concentrate CO_2 in air to a partial pressure of a few kPa, or create aqueous solutions of similar fugacity. The

CO_2 is then brought in contact with the rock material either as a gas or in an aqueous carbonate/bicarbonate solution. It is important that the water returned from the mineralization process is not brought in direct contact with the DAC system. If the DAC system inherently creates an aqueous solution, there is an "air gap" between the DAC product and the brine brought in contact with the minerals. This limits any harm that might be caused to DAC sorbents that come in contact with complex mineral brines. Two designs have been advanced for further analysis of the life cycle implications and the economic factors in design.

The system that emerges from the preliminary system design phase use CO_2 at partial pressures of 0.05 atm or less. This allows for the direct use of moisture and thermal swing sorbents without the need for additional process steps to further purify or concentrate the CO_2. Using such low concentrations of CO_2 prevents the formation of hydrated phases and clays in the product stream. Such hydration processes seem to prevent the formation of carbonates and have been observed both by us and other researchers (Mills *et al.*; Ehlmann, Berger, *et al.*; Ehlmann, Mustard, *et al.*). Reducing the risk of clay formation is a major design consideration.

Air Based System

We first envision a process where the heap is partially saturated with water and injected with a gas stream from below that has been enriched with CO_2 from some direct air capture system. The gas is piped into the bottom of the heap under a slight overpressure that results in it upward flow making contact with wet surfaces, which are kept wet by delivering water at or right below the top surface of the pile. This is a form brine-based mineralization as the CO_2 will dissolve into water layer covering the mineral surfaces and then react with the mineral. A possible process is described and documented here; however further study is needed. It is likely the reactor is slower than a process where brine is saturated with carbon dioxide and then introduced to the heap, and this system is likely to be more energy intensive to operate the necessary blowers to sustain movement of a gas stream through the heap.

The most basic process which could satisfy these requirements uses a reactor containing finely ground mafic rock through which gas flows near ambient pressure and 5% CO_2 flow. This process is shown if Fig. 5. Gas flow is maintained with a slight overpressure that is in part generated by CO_2 delivery from the DAC system. In analyzing the mineralization process, it is not necessary to specify the DAC system beyond that it needs to generate a gas stream of slightly elevated pressure with 5% CO_2. In most cases the balance of the gas is ambient air. The grinding process stream (4) operates on a dry mineral material that after grinding passes through a 75-micron sieve (200 mesh), and may require mixing of coarser grain material after grinding to achieve required permeability. In most implementations the moisture content of the material during carbonation must be controlled and depending on the porosity of the material could reach as high as 5 to 10% by weight. Brine is delivered near the top of the heap and allowed to trickle downward. Evaporation can be limited by covering the heap and introducing liquid below the cover.

The most straight forward implementation assumes that CO_2 can directly react with the mineral matrix and form carbonates. However, such processes are exceedingly slow and that in virtually all cases the carbonation reaction is greatly accelerated by the presence of water either in vapor form or as a liquid film. In the latter case, the expected reaction pathway of the CO_2 is to dissolve into water, contained in the pore space of the rock, where it then reacts to precipitate carbonate. The water naturally occurring in the rock may be insufficient to allow for this pathway to be realized and it will likely require the injection of additional water as liquid or possibly vapor along with the CO_2 into the heap.

A gas-solid reactor that takes advantage of aqueous films on mineral surfaces represents a hybrid between a solid-gas reaction system and a fully aqueous system described later in this report. While it is expected to improve on the reaction kinetics of a pure gas-based system, there are two mechanisms that may result in a slower reaction rate than for a fully aqueous system: firstly, the initial requirement for dissolution into the thin coating of water on the surface of the particles and secondly , the lack of convective transport of ions including silicates and carbonates away from the mineral particles. While many fully aqueous systems are also dependent on dissolving CO_2 into the brine, the surface area afforded by specifically designed processes for bringing CO_2 in contact with the brine may be much greater and thus allow for overall faster speeds. A lower speed of dissolution and the inability to transport ions away from the surface may result in a further decrease kinetics for the dissolution of CO_2. By preventing the circulation of ions away from the thin water film on the particles it may become difficult for the reactions to continue at the same rate as observed in fully aqueous system as the ions in the mineral rock are saturate the thin aqueous film and are no further leached out of the mineral matrix.

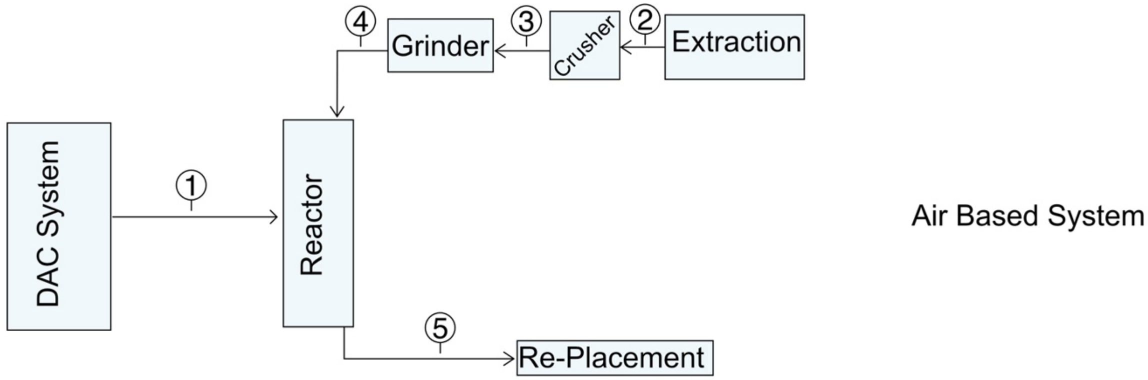

Figure 3: Air Based System Process Diagram.
The numbers marked in a circle with a downwardly extending flag are the process stream number and correspond to the stream number in table 1. Material flows into the reactor along streams 1 and 4 and leaves the reactor along stream 5 following the direction of the arrows.

Table 1: Air Based Stream Table
The process described is for a 1 ton per year CO_2 mineralization system. The reactor is the most significant piece of equipment and its size is a function of retention time. For every ten days of retention, a 1 ton per year reactor requires 1.1 m^3 of volume of the reactor assuming a 3% by mass uptake rate. At this phase of design, a 3% uptake at 30 days is taken as the baseline rate resulting in a reactor volume of 3.3m^3. This volume is substantially increased from those encountered for the fully aqueous system, to both account of the greater degree of uncertainty and expected decrease in reaction rates.

Air Based System Stream Table					
Process Stream Number	1	2	3	4	5
Material	5% CO2 Air	Extracted Rock	Crushed Rock	Ground Rock	Carbonated Rock
Flow Rate	$7.7*10^{-3}$ m^3s^{-1}	$1.1574*10^{-3}$ kg/s	$1.1574*10^{-3}$ kg/s	$1.2731*10^{-3}$ kg/s	$1.1921*10^{-3}$ kg/s
Pressure	1 ATM gage				
Particle Size		3cm < X < 0.5m	<3cm	<63 micron	<63 micron

Aqueous Based System

The second process design, while more complicated due to the need for three additional unit processes, is likely to be a more robust and ultimately more cost-effective system for further development. Use of an aqueous process is likely more efficient than an air-based system as the brine provides both a rapid delivery of carbonate ions for mineralization reactions and a means of transporting away dissolved reaction products including magnesium ions. While these advantages improve systems performance, they are not without cost due to the added complexity and for the need to minimize water loss in maintaining the brine in favorable ranges of salinity and preventing transport of water out of the system when disposing of the carbonated residue.

Management of water use was identified as a key factor for social and economic reasons that represents a major potential barrier to the implementation of *ex situ* mineralization technology. A major goal of this project was the development of approaches that can make use of currently economically unproductive sources of water including non-potable saline aquifers. The planned use of high salinity groundwater as the source of water for the project requires managing the buildup of salinity in the mineralization reactor. Salts enter the brine directly from the water supply and from non-carbonate salts leaching from the rock. Evaporation tends to concentrate these salts. Coupled with the need to manage salinity is the requirement for many DAC system to utilized steam, which conventionally is thought to require high purity feed stock water, or in the case of moisture swing sorbents requires high purity water. To achieve the level of purity required from saline aquifers will require water purification equipment which is energy intensive and costly or alternative approaches such as evaporation which generates clean water vapor, but at a significant energy cost. A third option is the direct use of the saline brine, but that requires managing a wastewater stream that is even more concentrated.

The social, political and legal aspects of water use represent perhaps the more important limiting constraint. While saline aquifers are non-potable, currently in Arizona, which is the focus of this project, they are treated by law as potable water supplies and are managed though the same social-legal framework that dominates water regulations in the western United States. This makes creating a new industry with potentially large and pervasive use of water a sensitive issue for local communities. Water management, and its benefits to communities must be addressed through both engineering methods reducing the need for water and through developing projects in a sensitive and responsible manner. Indeed, designs that produce fresh water as a byproduct will be of particular interest. Fresh water as a by-product could be designed into DAC and mineralization processes that start out from saline water sources. These considerations drove the need for implementing additional process units that directly affect water demand and water quality requirements.

The purpose of the first additional unit of operation is to dissolve the CO_2 into the carbonate brine. In a system with direct gas-solid interaction mediated by moist interfaces, this step happens implicitly inside the contactor. Gas-liquid interchanges at elevated CO_2 concentrations such as those anticipated here, allow for an increased rate of dissolution into the brine in comparison to the dissolution rate of CO_2 from an un-enriched air stream. Also, the maximum CO_2 fugacity that can be achieved is ultimately limited by the CO_2 partial pressure in the input gas stream. Such systems can take a variety of forms including water spray contactors or packings that maximize gas liquid interfaces. Alternatively, gas permeable membranes could allow for the efficient dissolution of CO_2 into the depleted brine. They would however come at a significant cost. Final selection depends on water efficiency and capital cost of the equipment. The challenge with any gas/liquid interface is the tendency to lose water by raising the moisture level in the exit gas stream.

The second added unit process is for salinity management in the form of evaporation ponds to allow for the discharge of some of the working brine after the salinity levels of non-carbonate salts increases to the point that it damages the overall reaction kinetics by significantly impacting the activity coefficients. Evaporation ponds while less efficient in terms of water loss per unit salinity discharged than a distillation or reverse osmosis system. Their use in these preliminary processes design aims to minimize capital costs by reducing equipment needs. Less and simpler equipment also reduces maintenance costs.

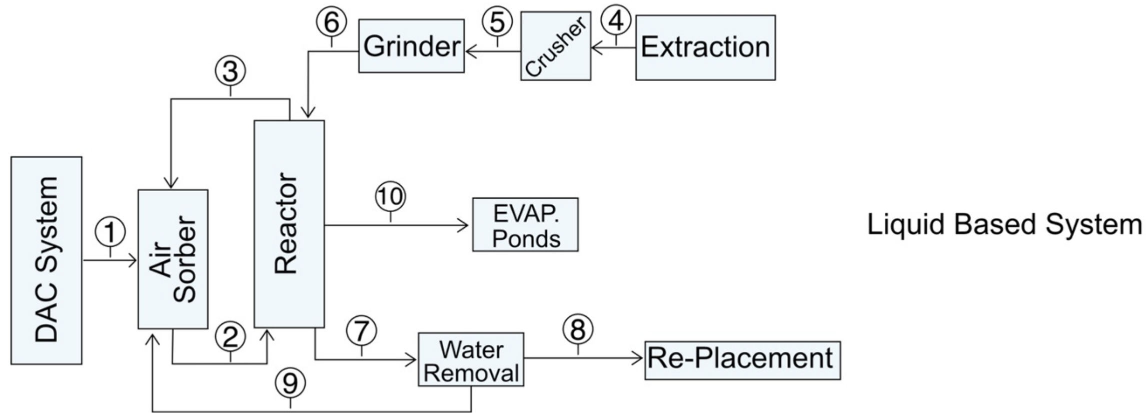

Figure 4: Liquid Based Process Model

Table 2: Liquid Based Stream Table
The process described is for a 1 ton per year CO_2 mineralization system. The reactor is the most significant piece of equipment and size is a function of retention time. For every ten days of retention a 1 ton per year requires 1.1 m^3 of volume of the reactor assuming a 3% by mass uptake rate, at this phase of design a 3% uptake at 10 days is taken as the baseline rate resulting in a reactor volume of 1.1m^3.

Table 2: Liquid Based System Stream Table										
Process Number	1	2	3	4	5	6	7	8	9	10
Material	5% CO2 Air	Brine	Brine	Extracted Rock	Crushed Rock	Ground Rock	Carbonated Rock Slurry	Carbonated Rock	Brine	Brine
Flow Rate	$7.7*10^{-3}$ m^3s^{-1}	4.0880 $*10^{-4}$ kg/s	4.0880 $*10^{-4}$ kg/s	1.1574 $*10^{-3}$ kg/s	1.1574 $*10^{-3}$ kg/s	1.2731 $*10^{-3}$ kg/s	2.7006 $*10^{-3}$ kg/s	1.286 $*10^{-3}$ kg/s	1.4146 $*10^{-3}$ kg/s	~0
Liquid Content	0%	100%	100%	0%	0%	0%	70%	10-20%	100%	100%
Pressure	0	4.1 ATM	0				0	0	0	0
Particle Size				3cm < X < 0.5m	<3cm	<63 micron	<63 micron	<63 micron		

Based on the experimental data developed for this project from mineral samples of deposits located in Arizona, we were able to construct a process flow that based on the modeling here should be feasible. We have been able to balance mass and energy flows and did not encounter bottlenecks or obstacles that would render the process impractical.

The initial concept of reverse heap leaching, i.e., using bicarbonate brines to drive the reaction in a simple pile of mineral resources proved to be viable and did not result in impractically long timescales. From an economic perspective it aims to avoid high capital cost equipment like pressure vessels and instead operate at time constants too long for typical technical equipment.

Based on the experimental data, the choice of mineral matters. Some raw materials were resistant to mineralization where as others from the same deposit reacted reasonable fast. This gives an advantage to small process modules that can use operational feedback to decide whether a charge of mineral rock is reactive or not. In the context of heap leaching this suggests long linear structures are preferable. Sections do not show reactivity will simply be left alone. It is worth noting that minerals whether they carbonate or not will have to be disposed of. A further refinement may use small samples to test the reactivity of a batch before it gets finely ground.

Since the process appears to be feasible, the next step in the analysis is to provide a first order TEA and LCA that then can be subsequently refined until one achieves practical economically viable process designs.

Process TEA

It is generally challenging to provide a reliable technoeconomic analysis of early-stage process designs (John Cirucci; Lackner and Azarabadi). Uncertainties arise because processes are not fully defined, equipment is not fully specified, and typically existing equipment is not well suited for the task, and the cost of modifications is difficult to estimate. Moreover, cost can vary dramatically over time as more equipment of a type is built. Appliances, computers, photovoltaic panels, and windmills have shown cost reductions by orders of magnitude. Such changes in cost are simply not predictable. Nevertheless, it is necessary to obtain some estimates of the cost and also the environmental footprint based on what is known today to make rational decisions to move forward with an approach and if so how to improve it.

One approach is to proceed by analogy. In other words, one is looking for similar processes used in other industries to obtain approximate bounds on cost. Going beyond such high-level analogies risks locking in specific technology and equipment choices that result in unnecessarily high costs. A second disadvantage of over-specifying the process is that it creates a false impression of accuracy where it does not exist. While the estimate of the costs for chosen equipment, its interconnections, and operations may be quite accurate, different requirements for this equipment or the potential for cheaper substitutes could make the result entirely irrelevant. The only benefit of such an approach that starts from a completely defined system and costs out all details is that it clearly sets an upper limit on the cost of an appropriate design. However, we believe it is premature to go this route; or put another way, we would expect the upper bound obtained in this manner to be too high to generate a useful constraint.

Our goal here is to figure out the range of costs one might expect these technologies to follow. Using an analogy may not guarantee an upper bound and be burdened with large uncertainties, but it generates a starting point for further refinement. The process can be strengthened by using multiple analogs and attempting to incorporate estimates of the cost differentials between the different analogs and the target technology. This approach to TEA will also help in determining the various contributors to the overall cost, identify bottlenecks in the supply chain and missing units of operation in the process design.

The LCA component of this work will look at the net carbon negativity of the sequestration activity. Since these processes consume energy and rely on complex supply chains with their own embedded carbon emissions the amount of carbon sequestered must be corrected for the indirect emissions unless a global-carbon-take-back obligation takes care of these emissions upfront. LCAs also will need to consider other environmental impacts. For these types of technologies, the impact of mining and tailing disposal and the environmental concerns related to impurities in the material, such as heavy metals or asbestos, would rank highly. The asbestos issue is known to be a problem for olivine rich serpentinite minerals. In the samples considered here, it appears not to be a problem. However heavy metals do occur in basalts, however in the samples we studied it was not a problem. In summary these materials were sulfur poor, contained minimal heavy trace elements, and no asbestos fibers. As with all mining operations care must be taken to prevent exposure of personnel and local communities to silica dust during and after mine operation.

Another purpose of TEA/LCA analysis is to consider combining the sequestration effort with generating beneficial products. It is unlikely that the fine particle size required for effective carbonation results in a product that could be utilized rather than disposed of. It is conceivable that the material can be rendered cementitious and could serve as a binder for other engineered materials. However, such markets do not exist and the technology readiness level of such applications would be extremely low, TRL 1-3, and are therefore they are no further considered. On the other hand, the necessity for fine grinding offers the opportunity to extract valuable components from the input stream prior to or after carbonation. One option is to pass the ground up material through a flotation or other separation process prior to heaping the material into a pile. Alternatively, the pile after it has been carbonated is once more moved through a flotation or similar process to extract useful compounds. A third option is the entire material goes through an extraction process prior to carbonation. Mafic rocks tend to be rich in certain metals like nickel, chromium, iron, and magnesium. The fact it has been ground up finely may also make it interesting as a source of rare earths and other elements that are not uncommon but expensive to separate from the matrix they are in.

A study was performed that used landfill disposal as an analog for the mineral sequestration processes considered here (Stirling).The basic concept called for an open pit mine with the material removed, the pit lined with clay, geomembrane, and geofabric for the leaching process to proceed. The pit construction was estimated based on EPA data on American land fill operations for municipal waste that performed logical similar tasks. Costs estimates therefore drew on costs known in the landfill industry. The study specified the grinding level to 1 micron, which suggests most particles are smaller than 1 micron. The reaction rates were informed by experimental results from this project, and resulted in time scales on the order of months (Sjolund *et al.*).

Based on this analysis the TEA by Rob Stirling generated estimates for the various cost components which resulted in a cost range from $128 to $279 dollars per tonne. Those costs according to the previous constraints are too high for economic viability, but warrant a more careful look at the details of the analysis and see whether further cost reductions are plausible. The first obvious question is whether the analogy to landfills is the most optimal. Heap leaching in mining operations from gold mining to [cheap option] could further inform the validity of this analogy. Beyond that, the report is valuable as it breaks out various cost components and therefore makes it possible to focus attention on the major contributors and see to what extent their contributions could be reduced.

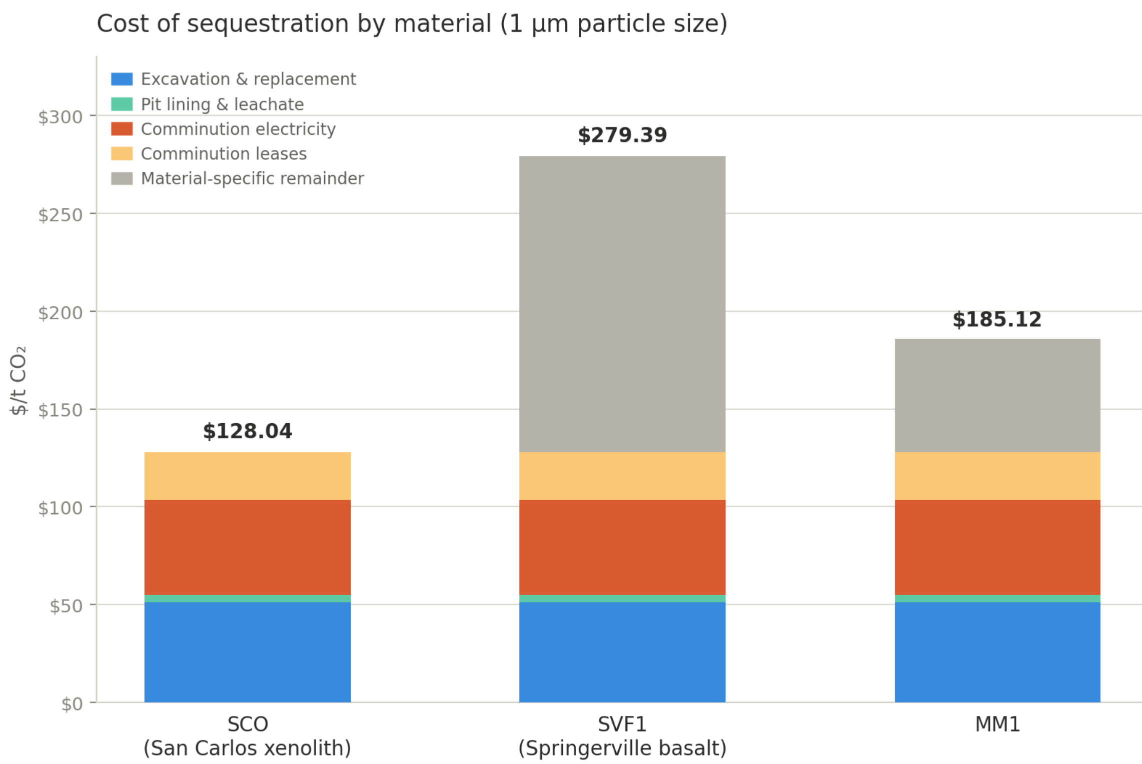

Figure 5: Lowest Cost Results from TEA prepared by Robert Stirling. Reproduced from Technical and Economic Evaluation of Semi In situ Mineral Carbonation.

In typical heap leaching processes for coper extraction, reported prices in dollar per ton of rock per year capacity for capital expenditures is found to be on the order of Dollars per tonne (Long *et al.* Table 1). Similarly operating costs in terms of dollars per tonne of rock can be calculated, with a range of $3.4/tonne to the most extreme case $16/tonne with prices centered around $4-$5 /tonne (Long *et al.* Table 1). This pricing is significantly lower than current pricing and indicates with improvements during design and implementation of the system significant cost savings may be achieved.

Another question that deserves further inquiry is the quality and durability of the barrier to prevent liquid losses from the mineral heap into the ground. The study by

Stirling provides a barrier designed for landfills. It not only prevents seepage during the operation it also is designed to last. Unlike in heap leaching in gold mines which deploy sodium cyanide or in coper mines using sulfuric acid, small leakages of sodium bicarbonate are far less concerning. Moreover, the concern about seepage effectively ends with the treatment operation. The heap itself could be piled directly on the ground without a barrier as it is essentially the same material as was there before. If anything, the carbonation process should result in a chemically more benign. Leachate from rainwater flowing through the end product should have a near neutral pH but not acidic enough to raise concerns of heavy metal leaching. The net result is that liners meant for landfills that have concerns over long term toxic leachates are substantially overengineered for this project. After completing the mineralization, it may be desirable to puncture any geomembranes in use to prevent ponding at the bottom of the pile. In any case a more thorough study would likely find substantial cost savings in relaxing the impermeability constraints of the barrier between the bottom of the pile and the ground below. Finally, it should be noted the cost of geomembrane taken from the EPA study suggest a cost of $40 /m^2, to which one needs to add the cost of a clay layer ($ /m$^{2)}$ and geo-composite ($ / m^2). Given that typical prices for geomembranes are between $1-$7 /m^2 the cost of an EPA regulated landfill may be higher than necessary for this application. If the leachate recovery system only has to survive for a few months of carbonation it is likely that the cost of its construction can be reduced.

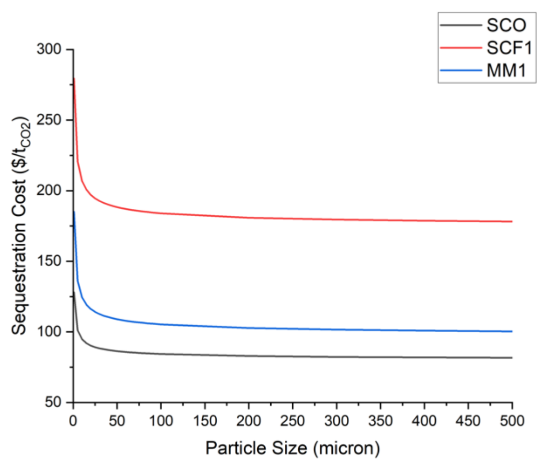

Figure 6: Total Sequestration Cost as Function of Particle Size prepared by Robert Stirling. Reproduced from Technical and Economic Evaluation of Semi In situ Mineral Carbonation .

Grinding to the #200-mesh level has average costs around $10-20 / tonne suggesting that it is more expensive than typical rock (Ballantyne and Powell). Both capital cost and operations cost will scale with particle size. As a rough rule of thumb, grinding to 1 micron is roughly an order of magnitude higher cost than grinding to 200 mesh, 75 micron. It is therefore critical to find the optimal between grinding costs and duration of the reverse heap leaching. Since the energy investment into moving fluids is very small, we would expect slowing down the process due to a larger particle size will likely improve economics.

It is clear form this analysis that many of the costs in this first attempt could be reduced by streamlining the operation. One place where the analysis may, however, be too optimistic is the assumption of relatively high conversion rates. It may be possible to bind up to 30% by weight to the rock material, but this has not been demonstrated in an industrial setting. It is evident from the study that the overall cost is quite sensitive to the total amount of rock that needs to be mined. On the other hand, we note, that the cost of fluid recirculation is nearly negligible and therefore could operate for extended periods of times to push the conversion as close to the theoretical maximum as possible.

Life Cycle Assessment

In performing a life cycle assessment, one will need to consider all components of the system. This includes the direct air capture component, the mineralization component which can be broken up into mining, comminution, and aqueous recycling of the carbonate brine to react with the rock. In addition, one will need to consider the energy source if it is part of the system. The direct air capture subsystem has a large life cycle footprint with regard to carbon accounting, water consumption, and energy consumption. Since it is not explicitly designed for this system, one will need to look at typical air capture LCA's for various designs. Most likely, off-the-shelf systems have a slightly higher footprint then those deployed here, as the mineralization system tends to relax technical specifications on the direct air capture. However, even under that assumption, the carbon and energy footprints of the DAC subsystem dominate the overall footprint, even if future DAC systems will likely improve. In any case, the nascent state of direct air capture injects a very large uncertainty into any life cycle assessment. In the economic analysis we simply assumed that DAC will eventually reach $100 / tonne. Here one might assume that energy consumption, will be better than currently best in class around 1000 kwh/tonne. The indirect carbon emissions of nearly every DAC system are directly tied to their energy consumption. They therefore depend on the source of energy ranging from a few grams per kWh for photovoltaic power, to a kg per kWh from coal fired power plants. Water consumption in direct air capture systems is impossible to predict without a detailed understanding of the specific DAC technology. Direct air capture systems run the gamut from collecting water from the atmosphere, to releasing 10-20 time the mass of water per mass of CO_2. Further complications arise from the source of water, which in some cases needs to be fresh water and in others could be saline brines. In the latter case, LCA would also have to consider the disposal of the more concentrated rejected brines.

From a carbon footprint perspective, the dominant concern for the mineralization process is its energy consumption which is mainly due to the energy demand for comminution. For high TRL levels the energy is likely to come from the grid. For more advanced implementations, that can manage the intermittency of renewable power that carbon footprint could be eliminated. Since the energy demand has been discussed in the TEA the LCA simply has to account for the carbon intensity of the energy source. Another concern for the LCA is the potential for water consumption in the carbonation process beyond direct air capture. In the technology discussed here, aqueous solutions are used which are prone to evaporate especially in the dry Arizona climate. One way to minimize this effect is to have the liquid released either below the surface of the heap, or to have the surface covered with geotextile. Either way evaporative losses can be greatly reduced. For now, this leaves us with a wide range of potential water losses. This uncertainty can only be reduced by studding practical implementations which are beyond the scope of this effort.

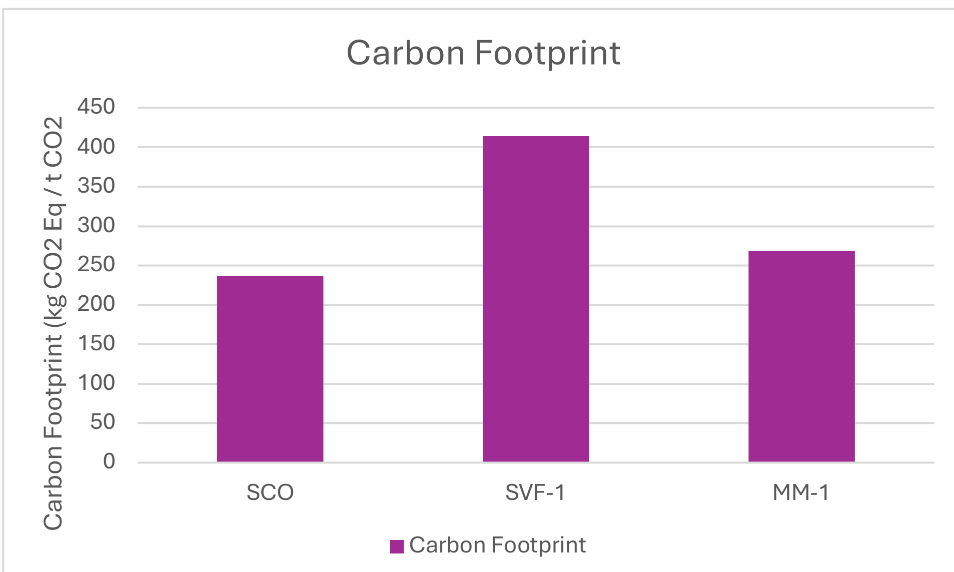

Figure 7: Carbon Footprint: Data for Inventory compiled from Terlouw et al. Elsing et al. and de Bortoli.

Land use, which is a frequently topic of life cycle assessments, has already been addressed in the TEA and sets the size of the standard operation at 50 acres. An LCA will also have to consider the impact of access roads, and how to partition these land use impacts among the mineral sequestration and potential coproducts. An item that has been ignored in the TEA is the need for land reclamation at the end of the mineralization process. Given the fact the material produced is not harmful, and can be easily covered with coarser untreated material, we expect this impact to be minor.

An issue that will have to be addressed in the future is the stability of the product against leaching of heavy metals and other impurities. At this point in time, the data for such a study is not available.

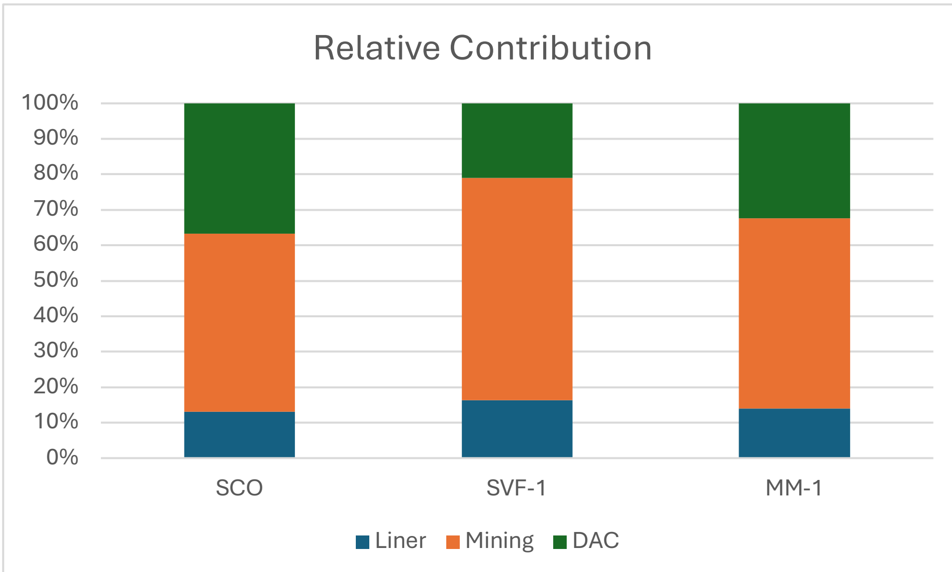

Figure 8: Relative Contribution of Process Activates. Data for Inventory compiled from Terlouw et al. Elsing et al. and de Bortoli.

Conclusion

Techno economic assessment and its further analysis reveal that off-the-shelf technologies will struggle to reach low costs, especially in mining operations that do not have the luxury of vast economies of scale. Nevertheless, there is reason for optimism. The low cost typically seen in large scale mines are in principle achievable in smaller systems as well, where economies of scale are replaced by economies of numbers. There are good reasons to believe that reverse heap leaching costs can be greatly reduced from that of the baseline TEA. The very conservative assumptions about the requirements of liners, and about the extremely small particle size can likely be relaxed. Most metallurgical processes operate successfully after milling to #200 mesh, 75 micron, and the minimal hazard of bicarbonate brines suggest significant cost reductions in these areas. If mass production reduces capital costs in a transition to solar electricity would result in intermittent or far cheaper energy costs. From an LCA perspective this removes the major obstacle of relying in carbon intensive energy. It does however greatly reduce the utilization rate of equipment.

A major obstacle to further cost reduction is the cost of rock mining. The amount of rock required critically depends on the calcium and magnesium content of the mineral source and the ability to utilize that content. In the Techno economic assessment mining costs were summarily delt with as a fixed cost at $15 / tonne of rock. This included extraction, transport, and replacement. In surface coal mining, the overburden removal extracts, transports, and replaces 10 to 20 times the mass of coal without raising the cost of coal to $150 or more. The freight on board price of powder river basin coal is below $16/ton (U.S. Energy Information Administration).

Despite lower target cost for mining, the cost of the mineral resource remains the most serious challenge to economic viability of *ex situ* mineral sequestration. This not only applies to the process considered here but as well to enhanced rock weathering or any other process that requires taking the mineral out of the ground. To give an example, at the costs of $3 /tonne for rock resources and a need of 3.3 tonnes of rock per tonne of CO_2 would result in mineral sequestration at $10 per tonne of CO_2 for the rock alone. At this extreme end, the cost is affordable. It is comparable to the cost of pipelining CO_2 over long distance which in other applications is the cost of operations. Another favorable cost comparison is to the compression and cleanup of CO_2 from a DAC system that produces wet CO_2 at one atmosphere to pipeline ready CO_2 $10 to $11 dollars per tonne (Abbas *et al.*). Many injection wells would require the same quality of CO_2. This last example suggests that under the right conditions mineral sequestration can compete with geological storage while providing a premium in accountability and long-term storage guarantees.

Even in the much more conservative TEA provided as part of this project, the costs of minerals with high carbonation potentials were only $50 / tonne of CO_2. While significantly less favorable it still provides an economically viable case. In practice we expect lower mining costs that, however, could be countered by lower carbonation potentials.

A path forward would require a confluence of favorable conditions. One should be looking for a resource with high carbonation potential and possibly a useful co-product like nickel, or chromium metal. Further research is needed to demonstrate that the metallurgical and carbonation processes can be performed on fairly coarse material. A tradeoff between finer grinding and longer reaction times will heavily lean towards slow process times. Finding locations that minimize the concern for groundwater contamination would reduce costs in lining the inverse leaching pits. Alternatively, one might try to combine carbonation with leaching of useful metals and thus raise the value of the operation. Access to water not necessarily of drinking water quality would be critical. Finally, the huge cost advantage of raw solar power suggests it is worthwhile developing an infrastructure that could be powered by intermittent solar voltaic. This however would violate our requirement of having already achieved a high TRL level.

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